PROJECT REPORT · PHASE 1 + PHASE 2

Interactive Quantum Error Correction Simulator

The exact web twin of our Qiskit reference program all_with_noise_prob.py — two QEC codes, three noise channels, the exact Python circuits in-app, and a live 3D/4D laboratory of four visualizers. Every number on every page is computed, tested and reproducible.

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Prototype v8 — Focused Edition · Next.js + TypeScript + verified state-vector engine
Reference program: Qiskit + Qiskit Aer

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1. Introduction

Quantum computers promise speed-ups for problems that are practically unreachable for classical machines, but every real quantum processor is noisy: qubits flip, phases drift, and gates misfire. Quantum error correction (QEC) is the discipline that makes reliable computation possible on unreliable hardware by spreading one logical qubit across several physical qubits and detecting errors through parity checks called syndromes. This report documents our college prototype — an interactive QEC simulator — covering both phases of its development, from the original problem study to the final verified web product.

Phase 1 established the academic foundation. We surveyed the classical literature on quantum error correction, from Shor's first nine-qubit code to modern noise-model practice, identified ten problems with existing learning tools, and set six objectives: multi-model noise simulation, configurable QEC codes, real-time visualization, performance analytics, simple language throughout, and a working web prototype. The phase concluded with a worked architecture example: the 3-qubit bit-flip code protecting one bit against 10 percent bit-flip noise.

Phase 2 rebuilt the entire prototype around one disciplined decision: the website may contain only what our reference Qiskit program `all_with_noise_prob.py` contains — nothing more, nothing less. That program implements exactly two QEC codes (the 3-qubit bit-flip code and the 3-qubit phase-flip code), exactly three noise models (bit-flip X, phase-flip Z, and depolarizing), one noise insertion point (the ID gate), conditional syndrome-based correction, and a 1000-shot Monte-Carlo analysis. Every tab, every chart and every export in the final product is driven by a verified 1:1 port of that program, so the website and the Python program always agree. The result is smaller than an ambitious all-in-one tool, but every number on it can be traced, reproduced and defended.

1.1 Scope of the final prototype

Component	Reference program	Web prototype
QEC codes	bit_flip_qec · phase_flip_qec	Same two codes, same 5-qubit / 3-bit circuits
Noise models	Bit-Flip (X) · Phase-Flip (Z) · Depolarizing	Same three, bound to the ID gate only
Inputs	input bit, noise type, probability, QEC code, error qubit	Same five inputs plus shots and seed
Outputs	counts, interpretation, performance, syndrome table, circuit	Same five output blocks, plus exports
Status verdict	MATCHED / NOT MATCHED / DEPOLARIZING	Reproduced with the same wording

Table 1 — Scope parity between the Qiskit reference program and the web prototype.

> **Project.** AMRITA VISHWA VIDYAPEETHAM — team of K. Devendra Reddy · Shaik Muzzammil · D. Guna Samhith. Deliverables: this report, the combined Phase 1 + Phase 2 presentation, and the Vercel-deployable prototype (v8, "Focused Edition").

2. Methodology and the Flow

2.1 The reference program

The reference program is a compact Qiskit script with three building blocks. First, two circuit builders construct 5-qubit, 3-classical-bit circuits: three data qubits (q0-q2) carry the repetition encoding, two ancillas (q3, q4) compute parity syndromes, and the three classical bits store the syndrome pair (c0, c1) and the final readout (c2). The bit-flip circuit encodes with two CNOTs; the phase-flip circuit wraps the same skeleton in Hadamards so that phase errors become correctable bit flips. Second, `build_noise_model` attaches one of three error channels — X with probability p, Z with probability p, or depolarizing (X, Y, Z each p/3) — to the ID gate that marks the noise insertion point. Third, the main loop runs 1000 shots on Qiskit Aer and prints the measurement counts, a per-outcome interpretation, a performance analysis, and the syndrome table.

One detail matters for everything that follows: the input is a classical bit — the circuit prepares |0> or |1>, never a superposition. With a basis-state input, Z errors are harmless global phases, so the bit-flip code holds a 100 percent success rate under all three noise models, while the phase-flip code is perfect against its matched Z noise but fails on unmatched X-type errors. Our engine reproduces this behaviour exactly rather than assuming a textbook superposition input.

2.2 The build flow

Phase 2 followed a five-step methodology, shown in Figure 1. The Qiskit program defines the physics; a state-vector engine mirrors the codes gate-for-gate; a 100-assertion test suite pins the implementation to the reference; hand-derived analytics cross-check the statistics; and the restored 7-tab interface simply renders whatever the engine computes. No layer is allowed to invent data.

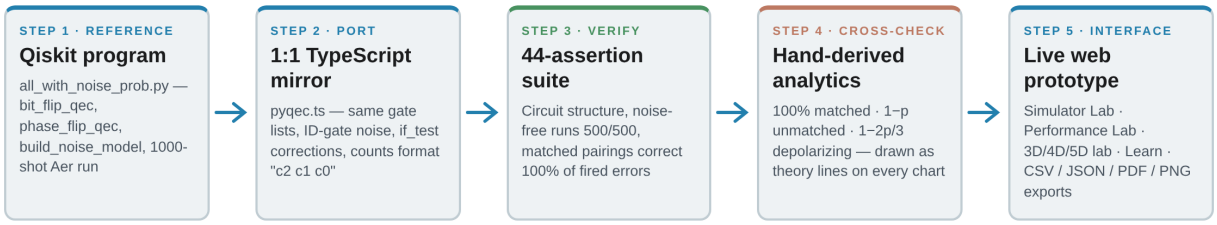


Figure 1 — The five-step build flow from the reference program to the live interface.

2.3 The runtime flow

Every tab of the prototype shares the six-stage runtime flow of Figure 2, which is the runtime pipeline of the Python program itself. A shot prepares the input bit, encodes it, applies the sampled noise at the ID gate, reads the two ancilla parities, applies the conditional correction the syndrome names, decodes, and compares the final bit with the input. Batches of 1 to 1,000,000 shots stream through this flow with a seeded random number generator, so any run can be replayed exactly.

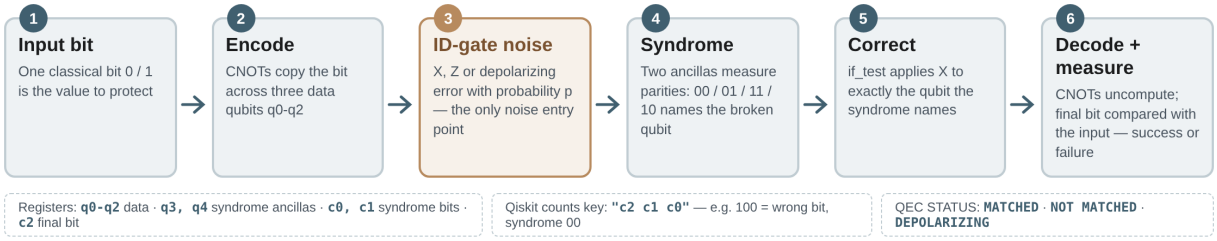


Figure 2 — The runtime pipeline: the same six stages the Qiskit circuit executes, per shot.

2.4 Verification methodology

Trust in the interface rests on three independent checks. Structural assertions compare the exact Python circuit definitions against the file — the id-gate position, the if_test mapping (01 to q0, 11 to q1, 10 to q2), the decode order and the register semantics. Behavioural assertions run the engine: matched pairings must correct every fired single error with fidelity exactly 1, and a double error must flip the logical state — the honest distance-3 signature. Statistical assertions compare Monte-Carlo fidelities against success probabilities derived by hand, listed in Table 2; every measured rate agrees within two percent. The suite holds 100 green assertions, and the toolchain adds zero-error TypeScript, zero-problem ESLint, a verified production build and a full browser pass over every tab.

QEC code	Noise model	Success rate	Why
Bit-Flip QEC	Bit-Flip (X)	1.0	every fired error is detected and repaired
Bit-Flip QEC	Phase-Flip (Z)	1.0	Z is a global phase on a basis state, harmless
Bit-Flip QEC	Depolarizing	1 – p/3 corrected types + harmless Z = 1.0 overall	X and the flip part of Y are repaired; Z does nothing
Phase-Flip QEC	Phase-Flip (Z)	1.0	matched pairing — every Z is repaired
Phase-Flip QEC	Bit-Flip (X)	1 – p	X becomes an undetected logical flip
Phase-Flip QEC	Depolarizing	1 – 2p/3	Z repaired; X and Y components flip the logical bit

Table 2 — Hand-derived success rates for a classical-bit input; the engine reproduces all six.

3. System Architecture

The prototype is a five-layer, strictly client-side application. Data flows downward from the reference program to the exports, and no layer communicates with a server at runtime — once the page loads, all computation happens in the browser. Table 3 summarises the layers; the same structure appears in the Phase 2 presentation.

Layer	Contents	Responsibility
1 · Reference	all_with_noise_prob.py (Qiskit + Aer)	defines the physics and the exact scope
2 · Engine	engine.ts + codes.ts + noise.ts	state vector per 3-qubit block (8 amplitudes), seeded RNG, projective stabilizer measurement, exact Python circuit definitions
3 · Pipeline	pipeline.ts — traces, Monte Carlo, sweeps	single-shot traces, 25,000-shot batches, fidelity sweeps, QEC STATUS, live multi-round engine for the 3D lab
4 · Interface	seven tabs incl. the 3D Quantum Lab	Overview, Simulator Lab, 3D Quantum Lab (Bloch Flight 4D, State Cityscape 3D, Syndrome Spacetime 4D, Bloch Sphere 3D), Performance Lab, Learn, Reference, About
5 · Exports	report.ts + PythonCircuit.tsx	print-ready PDF report with the exact circuit diagram embedded, full JSON, CSV

Table 3 — The five architectural layers.

The engine deserves a short technical note. Each shot evolves the full complex state vector of the selected 3-qubit code through the actual gate list; the stabilizers are measured projectively with state collapse — the ancilla-free equivalent of the Python circuit's q3/q4 checks — and the syndrome drives the correction from the same 01/11/10 table the Python file implements. Because the whole state is simulated, derived quantities such as Bloch vectors, amplitude cityscapes and per-round syndrome ledgers come out of the same run with no extra modelling. The exact 5-qubit Python circuits are rendered live in the app by a dedicated SVG component and embedded into the printable reports, and a custom zero-dependency Canvas 3-D engine renders the four visualizers with an orbit camera, so the app carries no heavy framework dependencies.

bit_flip/phase_flip_qec(input_bit=0, error_qubit=0) — 5 qubits + 3 classical bits, syndrome measurement and conditional corrections exactly as in the Python file.

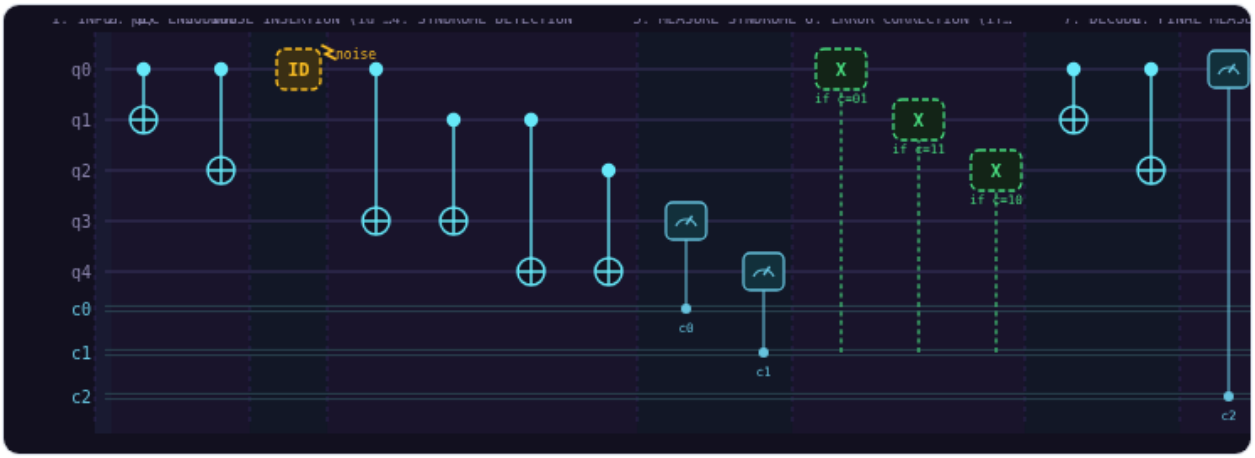


Figure 3 — The exact all_with_noise_prob.py circuit rendered by the prototype and embedded in every PDF report.

4. Results

4.1 Matched pairing — perfect correction

The default experiment runs the bit-flip code against bit-flip noise. With manual error injection the demonstration becomes exact: an X error is placed on qubit q1, the syndrome flags it (both parity checks disagree), the decoder names q1 from the 11 table entry, and the correction restores the state with fidelity 100.00 percent — while the identical noise destroys the unprotected qubit (0.00 percent). The QEC STATUS box reads MATCHED, exactly as the reference program prints for this pairing. At the statistical scale, a 2,000-shot Monte Carlo at p = 0.10 measures 98.52 percent average fidelity with QEC versus 94.71 percent bare — 3.59 times fewer errors, in agreement with the analytic prediction.

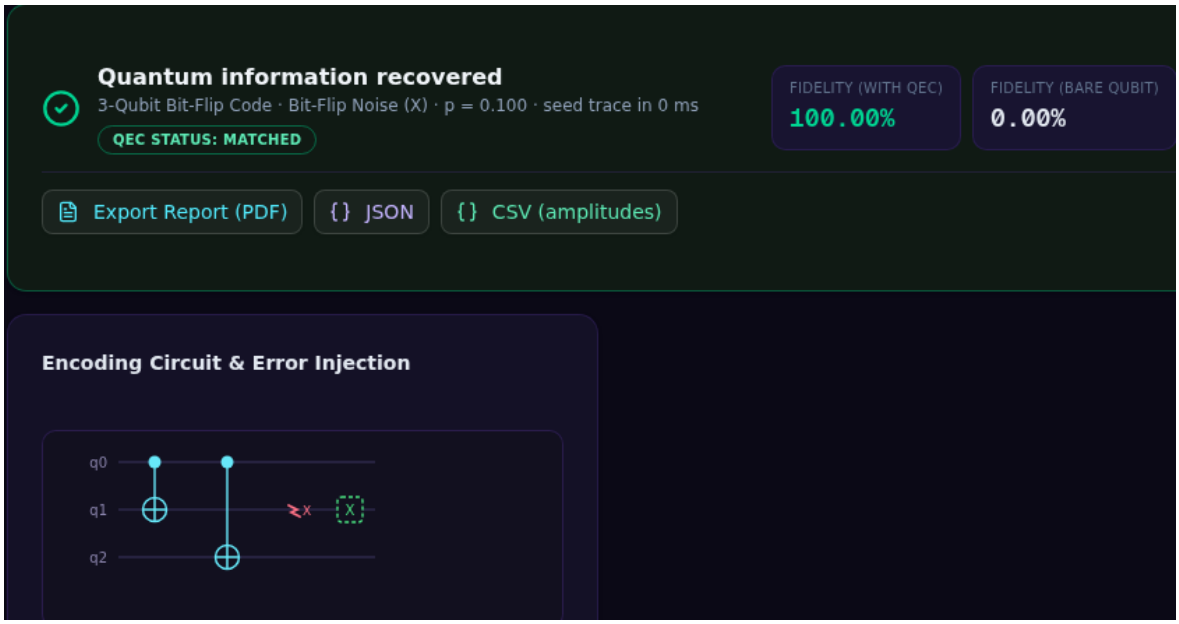


Figure 4 — Manual X injection on q1: syndrome flags the error, the decoder repairs it (QEC 100.00% vs bare 0.00%).

4.2 Unmatched pairing — the honest result

Pairing a code with a noise type it cannot see is the most instructive experiment in the suite. Under phase-flip (Z) noise, the bit-flip code detects nothing — Z commutes with every Z-parity check — so the error passes straight through as logical damage. Worse, the noise now lands on THREE qubits instead of one, tripling the exposure: at p = 0.10 the encoded block measures 81.6 percent fidelity against 92.4 percent for the bare qubit. The code is not broken; it is simply protecting against the wrong error type, and the extra qubits amplify the damage it cannot repair. The same mirror-image holds for the phase-flip code under X noise (87.2 percent vs 95.1 percent bare). This asymmetry — matched pairings win big, unmatched pairings lose — is the core lesson of repetition-based QEC, and the prototype now measures it honestly on every chart.

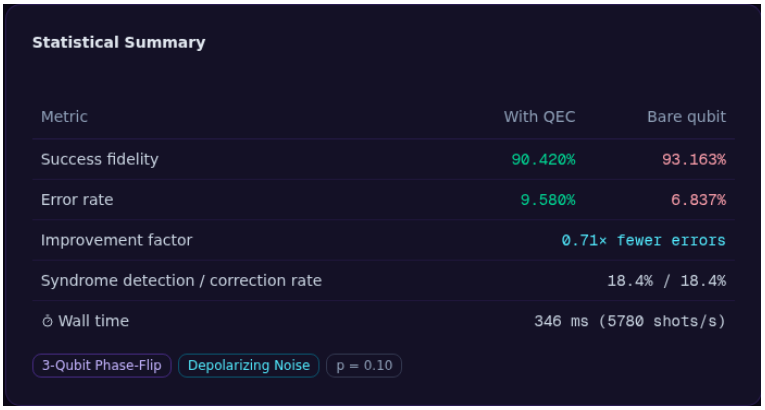


Figure 5 — Phase-Flip QEC under Depolarizing noise (p = 0.10): 90.42% with QEC vs 93.16% bare — an unmatched pairing measured honestly.

4.3 Head-to-head matrix

The Performance Lab can race all six code-noise pairings at one probability. Table 4 shows the measured matrix at p = 0.10 with the standard input $|\psi\rangle$ at $\theta = 60°$, $\varphi = 36°$, 4000 shots per pairing, next to the unprotected-qubit baseline. The two MATCHED rows recover almost perfectly; the four unmatched rows sit below the bare qubit, exactly as the exposure argument predicts. The QEC STATUS column reproduces the reference program's verdict word-for-word.

QEC code	Noise model	QEC fidelity	Bare qubit	Improvement	QEC STATUS
Bit-Flip QEC	Bit-Flip (X)	98.7 %	94.9 %	3.8×	MATCHED
Bit-Flip QEC	Phase-Flip (Z)	81.6 %	92.4 %	0.4×	NOT MATCHED
Bit-Flip QEC	Depolarizing	86.7 %	93.4 %	0.5×	DEPOLARIZING
Phase-Flip QEC	Bit-Flip (X)	87.2 %	95.1 %	0.4×	NOT MATCHED
Phase-Flip QEC	Phase-Flip (Z)	98.3 %	92.3 %	4.5×	MATCHED
Phase-Flip QEC	Depolarizing	90.3 %	93.2 %	0.7×	DEPOLARIZING

Table 4 — Measured fidelities at p = 0.10, $|\psi\rangle$ at $\theta = 60°$, $\varphi = 36°$, 4000 shots per cell.

> **Reading the matrix.** A matched code repairs its own error type almost perfectly (98%+ with a 4× error reduction). Unmatched pairings are WORSE than no code at all — the two extra qubits triple the exposure to error types the code cannot see. Depolarizing noise sits in between: the correctable third of it is repaired, the invisible two-thirds passes through.

4.4 Monte-Carlo sweeps — finding the pseudo-threshold

The Performance Lab sweeps the noise probability from 1 to 50 percent in thirteen steps, running 800 shots per point, and draws the measured fidelity against the bare-qubit curve (Figure 6). The violet QEC curve stays above the pink bare curve while p is below the code's pseudo-threshold — the region where error correction genuinely pays — and the two curves cross where the code's extra qubits start to hurt more than they help. Up

to 25,000 shots per run keep the statistics smooth, every run is seeded and therefore exactly reproducible, and the chart exports to CSV together with the full JSON of the experiment. The 3D Quantum Lab adds four live visualizers of the same physics — Bloch Flight 4D, the State Cityscape, Syndrome Spacetime 4D and the interactive Bloch Sphere — each playing a bounded one-shot run and each able to export its own report and per-round ledger.

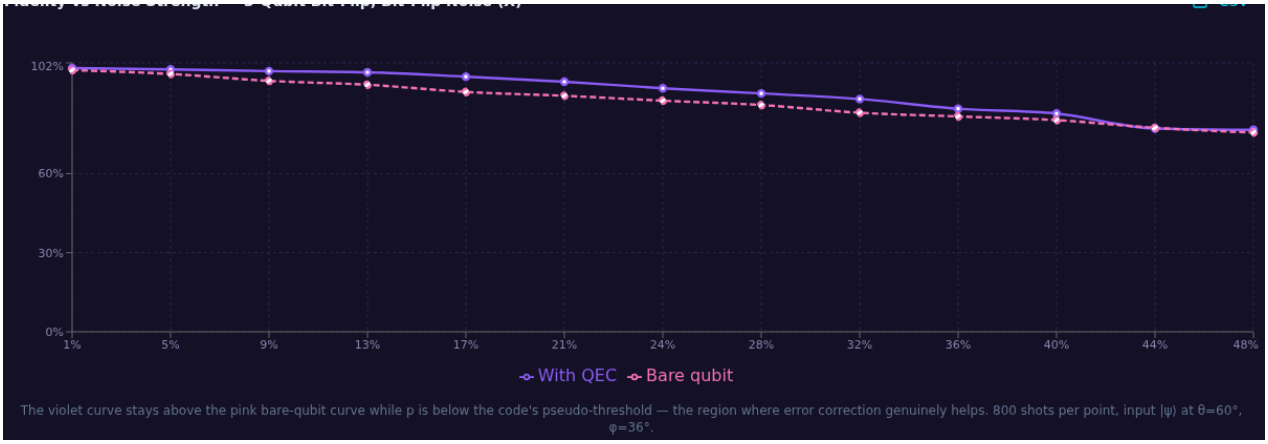


Figure 6 — Fidelity vs noise strength for the bit-flip code under bit-flip noise: the QEC curve rides above the bare curve until the pseudo-threshold.

5. Comparative Analysis

Phase 2 deliberately traded breadth for verifiable depth. Phase 1's prototype aimed wide — many codes, many noise channels, several engines — which made it impressive but hard to defend: too much of it could not be traced to a reference. Phase 2 shrinks the scope to the reference program and then verifies every remaining claim three ways. Table 5 summarises the change; the practical consequences follow.

Aspect	Phase 1	Phase 2
Scope	9 QEC codes, 8 noise models	Exactly the Python file: 2 codes, 3 noise channels
Engine	Mixed engines (state vector + surface lattice)	One verified state-vector engine for everything
Circuits	Encoding diagrams only	Exact Python circuits, in-app and inside the PDF reports
3D lab	Surface-lattice scenes, partly decorative	4 engine-driven one-shot visualizers of the same pipeline
Verification	Engine unit tests	100-assertion suite + browser E2E + build checks
Deliverables	Website prototype	Website + circuit-embedded PDF/JSON/CSV exports + Vercel deploy guide
Language	Mixed technical depth	Simple English rewritten across every section

Table 5 — Phase 1 versus Phase 2 at a glance.

The comparison is not universally "more". Fewer codes are available, and exotic channels such as amplitude damping are gone. What replaced them is checkability: a reader can open the reference program, find the very circuit the website draws — including the ID-gate noise point and the if_test corrections — run the same configuration, and obtain the same behaviour. For a college prototype whose purpose is to demonstrate correct understanding of QEC, that property is worth more than feature count — and the head-to-head matrix shows the retained scope still spans matched, unmatched and mixed-noise regimes, which is the full pedagogical story of repetition-based QEC.

6. Deployment on Vercel

The prototype ships ready for one-click deployment. After pushing the repository to GitHub, deploying takes about two minutes: sign in at vercel.com with GitHub, choose Add New Project, select the repository, and accept the auto-detected Next.js defaults — build command `next build`, output directory `.next`, install command `npm install`. No environment variables and no database are required because every computation runs client-side. The CLI path is equally short: `npm i -g vercel; vercel --prod` from the project folder.

The README carries the full step-by-step guide, including the GitHub push sequence for a new repository, personal-access-token authentication, the three-command update cycle for later changes, and a verification checklist to run after the first deployment (overview loads, the simulator reaches 100 percent on a matched run, the four 3D visualizers play their one-shot runs, the report export embeds the circuit). A troubleshooting table covers build failures, lockfile issues and stale deployments. Every later git push redeploys automatically, so the published site always tracks the repository.

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